

PRACTICE SET 1

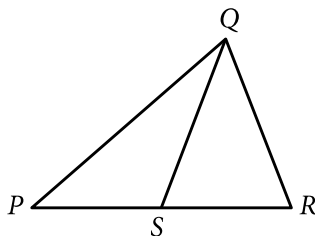
Quantitative Comparison Questions

For Questions 1 to 9, compare Quantity A and Quantity B, using additional information centered above the two quantities if such information is given. Select one of the following four answer choices and fill in the corresponding oval to the right of the question.

- (A) Quantity A is greater.
- (B) Quantity B is greater.
- (C) The two quantities are equal.
- (D) The relationship cannot be determined from the information given.

A symbol that appears more than once in a question has the same meaning throughout the question.

	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 1:	$(2)(6)$	$2 + 6$	(A) (B) (C) (D)



	<u>Quantity A</u>	<u>Quantity B</u>	<u>Correct Answer</u>
Example 2:	PS	SR	(A) (B) (C) (D)
			(since equal lengths cannot be assumed, even though PS and SR appear equal)

Points R , S , and T lie on a number line, where S is between R and T . The distance between R and S is 6, and the distance between R and T is 15.

	<u>Quantity A</u>	<u>Quantity B</u>	
1.	The distance between the midpoints of line segments RS and ST	The distance between S and T	(A) (B) (C) (D)

S is a set of 8 numbers, of which 4 are negative and 4 are positive.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|----------------------------------|-----------------|
| 2. | The average (arithmetic mean) of the numbers in S | The median of the numbers in S | (A) (B) (C) (D) |
-

The length of each side of rectangle R is an integer, and the area of R is 36.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|---|-------------------|-----------------|
| 3. | The number of possible values of the perimeter of R | 6 | (A) (B) (C) (D) |
-

$$x = (z - 1)^2$$

$$y = (z + 1)^2$$

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|--|-------------------|-----------------|
| 4. | The average (arithmetic mean) of x and y | z^2 | (A) (B) (C) (D) |
-

x , y , and z are the lengths of the sides of a triangle.

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|-------------------|-------------------|-----------------|
| 5. | $x + y + z$ | $2z$ | (A) (B) (C) (D) |
-

At a club meeting, there are 10 more club members than nonmembers. The number of club members at the meeting is c .

- | | <u>Quantity A</u> | <u>Quantity B</u> | |
|----|--|-------------------|-----------------|
| 6. | The total number of people at the club meeting | $2c - 10$ | (A) (B) (C) (D) |

n is a positive integer that is greater than 3 and has d positive divisors.

	<u>Quantity A</u>	<u>Quantity B</u>	
7.	n	2^{d-1}	(A) (B) (C) (D)

$$m = 10^{32} + 2$$

When m is divided by 11, the remainder is r .

	<u>Quantity A</u>	<u>Quantity B</u>	
8.	r	3	(A) (B) (C) (D)

$$xy = 8 \text{ and } x = y - 2.$$

	<u>Quantity A</u>	<u>Quantity B</u>	
9.	y	0	(A) (B) (C) (D)

Multiple-choice Questions—Select One Answer Choice

For Questions 10 to 17, select a single answer choice.

10. The area of circle W is 16π and the area of circle Z is 4π . What is the ratio of the circumference of W to the circumference of Z ?
- Ⓐ 2 to 1
Ⓑ 4 to 1
Ⓒ 8 to 1
Ⓓ 16 to 1
Ⓔ 32 to 1
-
11. In the xy -plane, a quadrilateral has vertices at $(-1, 4)$, $(7, 4)$, $(7, -5)$, and $(-1, -5)$. What is the perimeter of the quadrilateral?
- Ⓐ 17
Ⓑ 18
Ⓒ 19
Ⓓ 32
Ⓔ 34
-

DISTRIBUTION OF THE
HEIGHTS OF 80 STUDENTS

Height (centimeters)	Number of Students
140–144	6
145–149	26
150–154	32
155–159	12
160–164	4
Total	80

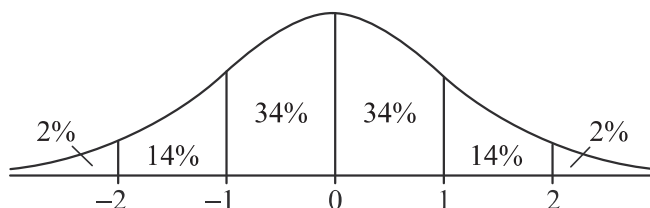
12. The table above shows the frequency distribution of the heights of 80 students, where the heights are recorded to the nearest centimeter. What is the least possible range of the recorded heights of the 80 students?
- Ⓐ 15
Ⓑ 16
Ⓒ 20
Ⓓ 24
Ⓔ 28

13. Which of the following functions f defined for all numbers x has the property that $f(-x) = -f(x)$ for all numbers x ?

- (A) $f(x) = \frac{x^3}{x^2 + 1}$
 (B) $f(x) = \frac{x^2 - 1}{x^2 + 1}$
 (C) $f(x) = x^2(x^2 - 1)$
 (D) $f(x) = x(x^3 - 1)$
 (E) $f(x) = x^2(x^3 - 1)$

14. If 10^x equals 0.1 percent of 10^y , where x and y are integers, which of the following must be true?

- (A) $y = x + 2$
 (B) $y = x + 3$
 (C) $x = y + 3$
 (D) $y = 1,000x$
 (E) $x = 1,000y$



15. The figure above shows the standard normal distribution, with mean 0 and standard deviation 1, including approximate percents of the distribution corresponding to the six regions shown.

The random variable Y is normally distributed with a mean of 470, and the value $Y = 340$ is at the 15th percentile of the distribution. Of the following, which is the best estimate of the standard deviation of the distribution?

- (A) 125
 (B) 135
 (C) 145
 (D) 155
 (E) 165

16. A car dealer received a shipment of cars, half of which were black, with the remainder consisting of equal numbers of blue, silver, and white cars. During the next month, 70 percent of the black cars, 80 percent of the blue cars, 30 percent of the silver cars, and 40 percent of the white cars were sold. What percent of the cars in the shipment were sold during that month?
- Ⓐ 36%
- Ⓑ 50%
- Ⓒ 55%
- Ⓓ 60%
- Ⓔ 72%
17. If an investment of P dollars is made today and the value of the investment doubles every 7 years, what will be the value of the investment, in dollars, 28 years from today?
- Ⓐ $8P^4$
- Ⓑ P^4
- Ⓒ $16P$
- Ⓓ $8P$
- Ⓔ $4P$

Multiple-choice Questions—Select One or More Answer Choices

For Questions 18 to 19, select all the answer choices that apply.

18. The distribution of the numbers of hours that students at a certain college studied for final exams has a mean of 12 hours and a standard deviation of 3 hours. Which of the following numbers of hours are within 2 standard deviations of the mean of the distribution?

Indicate all such numbers.

- ☐ A 2
- ☐ B 5
- ☐ C 10
- ☐ D 14
- ☐ E 16
- ☐ F 20

19. In a certain sequence of numbers, each term after the first term is found by multiplying the preceding term by 2 and then subtracting 3 from the product. If the 4th term in the sequence is 19, which of the following numbers are in the sequence?

Indicate all such numbers.

- ☐ A 5
- ☐ B 8
- ☐ C 11
- ☐ D 16
- ☐ E 22
- ☐ F 35

Numeric Entry Questions

For Questions 20 to 21, enter your answer in the answer box(es) below the question.

- Your answer may be an integer, a decimal, or a fraction, and it may be negative.
- If a question asks for a fraction, there will be two boxes—one for the numerator and one for the denominator. A decimal point cannot be used in a fraction.
- Equivalent forms of the correct answer, such as 2.5 and 2.50, are all correct. Fractions do not need to be reduced to lowest terms, though you may need to reduce your fraction to fit in the boxes.
- Enter the exact answer unless the question asks you to round your answer.

20. In a single line of people waiting to purchase tickets for a movie, there are currently 10 people behind Shandra. If 3 of the people who are currently in line ahead of Shandra purchase tickets and leave the line, and no one else leaves the line, there will be 8 people ahead of Shandra in line. How many people are in the line currently?

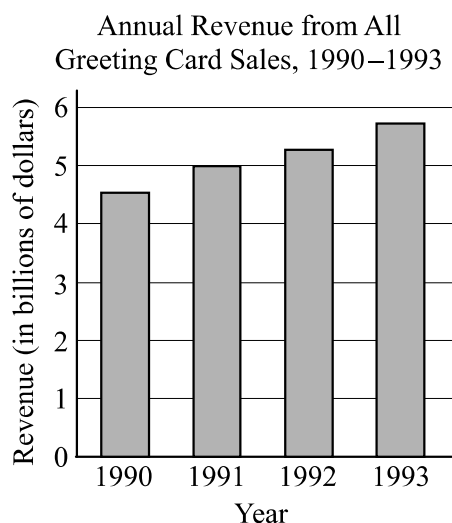
people

21. When the decimal point of a certain positive decimal number is moved six places to the right, the resulting number is 9 times the reciprocal of the original number. What is the original number?

Data Interpretation Set

Questions 22 to 25 are based on the following data. For these questions, select a single answer choice unless otherwise directed.

SELECTED DATA FOR GREETING CARD SALES



Number of Greeting Cards Sold for Ten Occasions in 1993

Occasion	Number of Cards
Christmas	2.4 billion
Valentine's Day	900 million
Easter	158 million
Mother's Day	155 million
Father's Day	102 million
Graduation	81 million
Thanksgiving	42 million
Halloween	32 million
St. Patrick's Day	18 million
Jewish New Year	12 million
Total	3.9 billion

Note: 1 billion = 1,000,000,000

22. In 1993 the number of Valentine's Day cards sold was approximately how many times the number of Thanksgiving cards sold?
- (A) 20
(B) 30
(C) 40
(D) 50
(E) 60
23. In 1993 a card company that sold 40 percent of the Mother's Day cards that year priced its cards for that occasion between \$1.00 and \$8.00 each. If the revenue from sales of the company's Mother's Day cards in 1993 was r million dollars, which of the following indicates all possible values of r ?
- (A) $155 < r < 1,240$
(B) $93 < r < 496$
(C) $93 < r < 326$
(D) $62 < r < 744$
(E) $62 < r < 496$

24. Approximately what was the percent increase in the annual revenue from all greeting card sales from 1990 to 1993 ?
- (A) 50%
 - (B) 45%
 - (C) 39%
 - (D) 28%
 - (E) 20%

For Question 25, select all the answer choices that apply.

25. In 1993 the average (arithmetic mean) price per card for all greeting cards sold was \$1.25. For which of the following occasions was the number of cards sold in 1993 less than the total number of cards sold that year for occasions other than the ten occasions shown?

Indicate all such occasions.

- ☐ A Christmas
- ☐ B Valentine's Day
- ☐ C Easter
- ☐ D Mother's Day
- ☐ E Father's Day
- ☐ F Graduation
- ☐ G Thanksgiving
- ☐ H Halloween